RATING METHODOLOGY

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Rating Methodology

Restaurants

This rating methodology replaces the *Restaurants* methodology published on 5 August 2021. While this methodology reflects the same core principles as the 2021 methodology, we have updated the scorecard to use linear interpolation in the scoring of quantitative factors and subfactors. We have also made editorial changes throughout the methodology to enhance readability, including expanding the "Other considerations" section for enhanced consistency across our corporate sector methodologies.

Scope

This rating methodology applies to companies globally that are primarily engaged in owning, operating or franchising restaurants. Restaurant companies that we rate using this methodology include: (i) companies that predominantly own and operate restaurants; (ii) companies that predominantly franchise restaurants; and (iii) companies that both own and franchise restaurants.

The determination of a company's primary business is generally based on the preponderance of the company's business risks, which are usually proportionate to the company's revenue, earnings and cash flows.

Sector overview

Restaurant companies pursue different business models, which influences credit quality because the level of earnings volatility, capital and investment intensity, margins and asset coverage vary in each model. In this section, we describe the main business models for illustrative purposes.

Many companies in the restaurant sector fit within one of three business models

Pure franchise-based model

» Low absolute level of revenue and operating profit – With franchisee fees and royalty rates in the low-to-mid-single-digit percentage range, the absolute level of revenues is significantly smaller than systemwide operations would indicate.

» High operating margins – With minimal, if any, costs associated with the royalty or franchise fee the operating margins are exceptionally high.

» Low capital expenditure requirements – The absence of any ownership interest in the actual restaurants eliminates the need for any capital expenditures for maintenance, refurbishment, or new unit expansion, which is all borne by the franchisee or affiliate base.

» Revenue stream is less volatile – Since the revenue stream is based on a percentage of a franchisee's gross revenues and not its earnings, operating performance associated with a franchised-based model tends to be less volatile.

» Low tangible asset value – The non-owner/operator model does not require any investment in restaurants or equipment and as a result there is no significant tangible asset value associated with the pure franchise model nor is there the opportunity to generate alternate liquidity from re-franchising units.

Restaurant owner/operator model

» Revenue generation and operating profit is reflective of all systemwide sales and as a result is considerably larger versus a franchisee-based model which only represents a small percentage.

» Lower margins – Despite the higher level of absolute operating income the company incurs all costs associated with operating the individual restaurants and as a result margins are more reflective of this higher cost structure.

» Capital intensive – The owner/operator bears the burden of all capex required to support a base of restaurants including ongoing unit maintenance, refurbishment, rebuilds, and new unit expansion.

» Asset value – Owner/operator models tend to have a mixture of owned and leased property and equipment which provide a level of tangible asset coverage. They also usually have some alternate sources of liquidity through the re-franchising of owned units.

» Higher level of volatility – This model is fully exposed to the gyrations of consumer habits and the impact of commodity cycles.

Combination model

» In most cases, companies in the quick service restaurant (QSR) sector and fast casual sector use a combination of both models with a heavier focus on franchised restaurants.

» Most of the large QSRs have adopted a business model in which they are targeting ownership of total systemwide restaurants at under 25% and some much lower than this.

» In the case of casual dining companies, the majority of units are owned although a number of companies have recently changed this practice and have begun re-franchising programs albeit the percentage is much smaller than QSRs.

General characteristics common across the restaurant industry

» There are several characteristics that apply to the majority of restaurant companies, although there are always some exceptions.

» Minimal working capital requirements – Given the cash-based nature of sales and the perishable nature of inventory, working capital requirements are minimal for most companies. With the majority of transactions between the consumer and restaurant settled with either cash or credit card, the change in a company's account receivable balance is generally modest. Inventory changes are also marginal due to the perishable nature of most inventory items.

» The most common industry practice is to use operating leases rather than ownership of real estate, which may result in a standard adjustment to recognize the debt equivalent of leases. Please see our cross-sector methodology that discusses financial-statement adjustments for nonfinancial corporations.[1]

» Industry measures of performance:

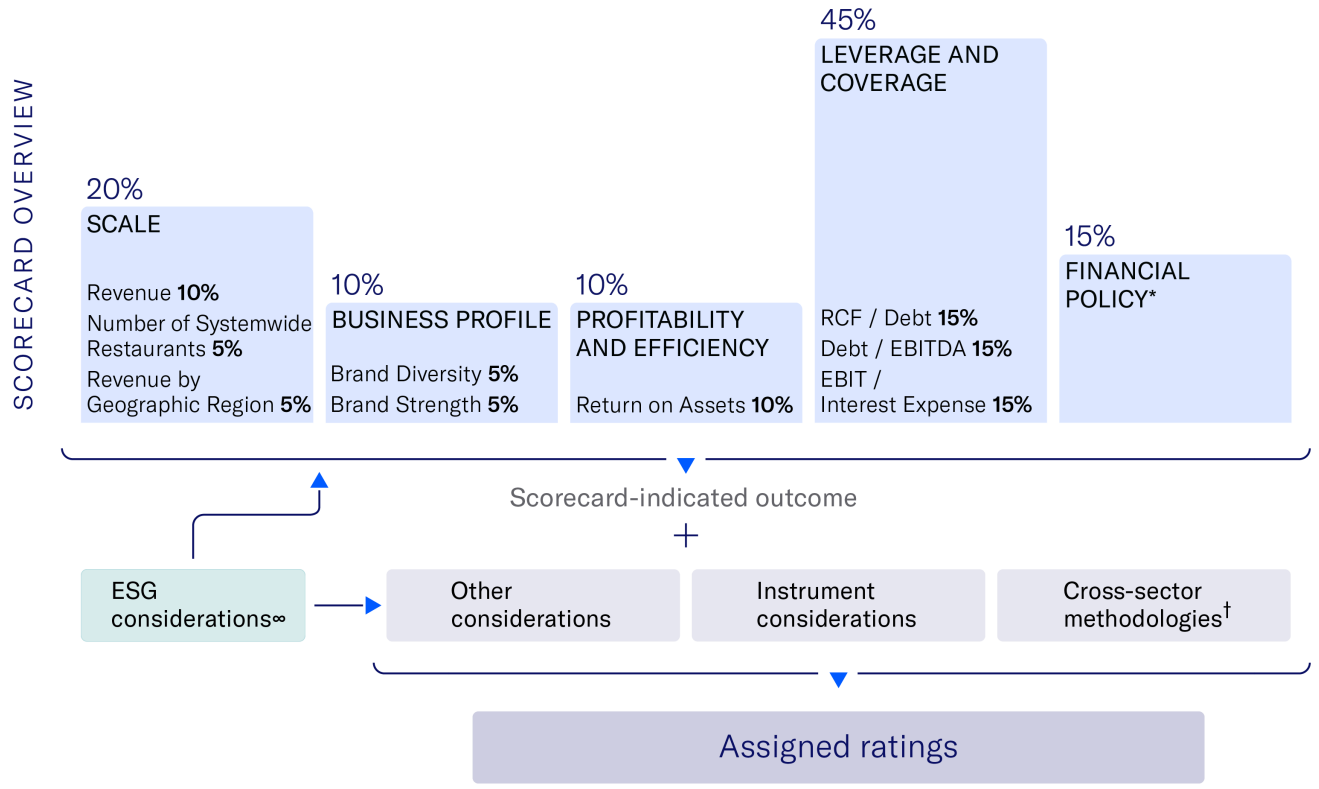
- **Same store sales:** Calculated as the percentage change in sales of those restaurants opened for more than thirteen months and are comprised of the percentage change in average check, product mix, and traffic or transactions. This industry standard is one of the key measures of a company's operating performance although the components are more telling than the aggregate same-store sales percentage change.
- **Average unit volumes (AUV):** A standard industry measure of performance although the bands surrounding AUV and the types of units incorporated are also informative.
- **Restaurant margins:** An indicator of operating performance at the restaurant level before the allocation of corporate overhead and other non-restaurant specific costs.

Rating approach

In this methodology, we explain our general approach to assessing credit risks of companies in this sector, including the quantitative and qualitative factors likely to affect rating outcomes.

The following exhibit illustrates our overall rating approach in the analysis of companies in this sector, which includes the use of a scorecard. The scorecard-indicated outcome may not match a company's assigned rating, which is determined by a rating committee. For more information, see the "Other considerations" and "Limitations" sections.

Exhibit 1
Illustration of the restaurants rating approach



* This factor has no subfactors.
∞ Environmental, social and governance (ESG) considerations, including, where available, our opinions of exposure to them as expressed in Issuer Profile Scores (IPSs), may affect scorecard factors and other considerations outside of the scorecard. For more information, see the "Other considerations" section.
† Some of the methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.
Source: Moody's Ratings

Scorecard

The scorecard in this methodology is a relatively simple reference tool that can be used in most cases to approximate credit profiles of companies in this sector and to explain, in summary form, many of the factors that are generally most important in assigning ratings. The scorecard may not include or address every consideration a rating committee may assess in assigning ratings to companies in this sector.

Exhibit 2

Restaurants scorecard

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Factor: Scale (20%)									
Revenue (USD Billion) [1]	10%	≥ $40	$23 - $40	$11 - $23	$5 - $11	$2.25 - $5	$0.5 - $2.25	$0.25 - $0.5	< $0.25
Number of Systemwide Restaurants [2]	5%	≥ 55,000	30,000 - 55,000	15,000 - 30,000	5,000 - 15,000	1,500 - 5,000	400 - 1,500	100 - 400	< 100
Revenue by Geographic Region	5%	Highly international with no single continent representing a majority of revenues.	No single country represents a majority of revenues.	A single country represents the majority of revenues while international operations are substantial.	A single country represents the substantial majority of revenues although international operations are material. Should have full national coverage.	A single country represents the substantial majority of revenues although international operations are meaningful.	The majority of restaurants reside within multiple regions in a single country.	The majority of restaurants are widely located within a single region.	The majority of restaurants are narrowly located within a portion of a single region.
Factor: Business Profile (10%)									
Brand Diversity	5%	Multiple brands (> 3) each with significant scale, equal in weight, differentiated product offering; day-part divisions that are equally weighted through all periods; and varied menu offering.	Multiple concepts (at least 3) each with significant scale, equal weight, and differentiated product offerings as well as day-part divisions that are more weighted towards two periods and menu offering that is varied; or a single brand with a leading market position within the industry, well diversified day-part throughout all periods and a varied menu offering.	Multiple concepts (at least 3) each with sizeable scale, differentiated product offering, and relatively equal weight; or a single brand with a leading market position within the industry, day part division that covers most day parts but is more weighted towards three periods but not equally weighted, and a menu offering that is varied.	Multiple concepts (at least 2) each with sizeable scale, differentiated product offerings, but relatively un-equal weight; or a single brand with a leading market position within the industry, multiple day-parts equally weighted towards at least two periods, and menu offering that is varied.	Multiple concepts (at least 2) each with adequate scale and differentiated product offerings, although un-equal weight; or a single brand with a leading market position in its respective category, with multiple day-parts more weighted towards two periods, and menu offering that is varied.	Could have multiple concepts although a single brand with a competitive position within its respective category is the dominant driver of the business with multiple but un-equal day-parts focused more towards two periods with a menu offering that is varied.	Single concept with a limited product offering and concentrated day-part.	Single concept with narrow product offering and single day part.

	Weight	Aaa	Aa	A	Baa	Ba	B	Caa	Ca
Brand Strength	5%	Expected to have solid same store sales ("SSS") performance at all concepts, driven equally by regular increases in average check and steadily increasing traffic, along with a consistently strong product pipeline that have always resulted in successful new product offerings.	Expected to have positive SSS at all concepts, driven by both average check and traffic, along with a strong product pipeline that have consistently resulted in successful new product offerings. Expected to have modest volatility across economic cycles.	Expected to have positive SSS at the majority of concepts, driven by both average check and traffic, along with a strong product pipeline that have resulted in successful new product offerings. Expected to have modest volatility across economic cycles.	Positive SSS at the majority of concepts with average check being the primary driver, although traffic remains consistently positive. A competitive product pipeline that yields good success with new offerings. Expected to have modest volatility across economic cycles.	Positive SSS although traffic patterns are more volatile and can migrate into negative territory for periods of time. Competitive new product pipeline that yields relatively good success with new product offerings.	SSS are more volatile with weak traffic patterns that mitigate pricing power and result in relatively weak market metrics overall. The product pipeline is adequate but success of new product offerings is mixed.	SSS are weak with negative traffic patterns for extended periods eliminating pricing power. Weak product pipeline and limited success with new offerings.	Extended period of negative SSS with no expected recovery in the near term. There has been a significant decline in the number of systemwide restaurants.
Factor: Profitability and Efficiency (10%)									
ROA (NPATBUI / Avg. Assets) [3]	10%	≥ 15%	11% - 15%	7.5% - 11%	5% - 7.5%	2.5% - 5%	1% - 2.5%	0% - 1%	< 0%
Factor: Leverage and Coverage (45%)									
RCF / Debt [4]	15%	≥ 55%	45% - 55%	35% - 45%	25% - 35%	15% - 25%	5% - 15%	0% - 5%	< 0%
Debt / EBITDA [5]	15%	≤ 1x	1x - 2x	2x - 3x	3x - 4x	4x - 5x	5x - 6.5x	6.5x - 8x	> 8x
EBIT / Interest Expense [6]	15%	≥ 12x	8x - 12x	5x - 8x	3x - 5x	2x - 3x	1x - 2x	0.5x - 1x	< 0.5x
Factor: Financial Policy (15%)									
Financial Policy	15%	Expected to have extremely conservative financial policies (including risk and liquidity management); very stable metrics; essentially no event risk that would cause a rating transition; and public commitment to a very strong credit profile over the long term.	Expected to have very conservative financial policies (including risk and liquidity management); stable metrics; minimal event risk that would cause a rating transition; and public commitment to a strong credit profile over the long term.	Expected to have predictable financial policies (including risk and liquidity management) that preserve creditor interests; although modest event risk exists, the effect on leverage is likely to be small and temporary; strong commitment to a solid credit profile.	Expected to have financial policies (including risk and liquidity management) that balance the interests of creditors and shareholders; some risk that debt-funded acquisitions or shareholder distributions could lead to a weaker credit profile.	Expected to have financial policies (including risk and liquidity management) that tend to favor shareholders over creditors; above-average financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that favor shareholders over creditors; high financial risk resulting from shareholder distributions, acquisitions or other significant capital structure changes.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring in varied economic environments.	Expected to have financial policies (including risk and liquidity management) that create elevated risk of debt restructuring even in healthy economic environments.

[1] For the linear scoring scale, the Aaa endpoint value is $60 billion. A value of $60 billion or better equates to a numeric score of 0.5. The Ca endpoint value is zero. A value of zero equates to a numeric score of 20.5.
[2] For the linear scoring scale, the Aaa endpoint value is 90,000. A value of 90,000 or better equates to a numeric score of 0.5. The Ca endpoint value is 0. A value of 0 equates to a numeric score of 20.5.
[3] For the linear scoring scale, the Aaa endpoint value is 20%. A value of 20% or better equates to a numeric score of 0.5. The Ca endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 20.5.
[4] For the linear scoring scale, the Aaa endpoint value is 100%. A value of 100% or better equates to a numeric score of 0.5. The Ca endpoint value is (5)%. A value of (5)% or worse equates to a numeric score of 20.5.
[5] For the linear scoring scale, the Aaa endpoint value is 0x. A value of 0x equates to a numeric score of 0.5. The Ca endpoint value is 12x. A value of 12x or worse equates to a numeric score of 20.5, as does a negative Debt/EBITDA value.
[6] For the linear scoring scale, the Aaa endpoint value is 20x. A value of 20x or better equates to a numeric score of 0.5. The Ca endpoint value is 0x. A value of 0x or worse equates to a numeric score of 20.5.
Source: Moody's Ratings

Discussion of the scorecard factors

In this section, we explain why each scorecard factor or subfactor is a meaningful credit indicator, and we discuss our analytical approach to scoring each factor or subfactor. When a factor comprises subfactors, we score at the subfactor level. Some factors do not have subfactors, in which case we score at the factor level.

In assessing quantitative subfactors, we use information that we calculate or estimate from a company's financial statements or regulatory filings, or that we derive from other observations. We may incorporate nonpublic information qualitatively in our assessment.

Our ratings are forward-looking and reflect our expectations for future financial and operating performance. However, historical results are helpful in understanding patterns and trends in a company's performance, as well as for peer comparisons. We typically calculate or estimate financial metrics, unless otherwise indicated, based on an annual or other 12-month period. However, we may assess the scorecard factors using various periods. For example, we may find it analytically useful to examine both historical and expected future performance for periods of several years or more.

All of the quantitative credit metrics in the scorecard incorporate our standard adjustments to income statement, cash flow statement and balance sheet amounts for items described in our cross-sector methodology on financial statement adjustments for nonfinancial corporations.[2] We may also make nonstandard adjustments that are specific to a particular company.

For a discussion of scorecard mechanics, see the "Scorecard-indicated outcome" section.

Factor: Scale (20% weight)

Why it matters

Scale is an important indicator of the overall strength of a company's business, the size of its market and the company's relevance within an industry and the economy. Scale is also an indicator of a company's importance to its customers and bargaining strength with suppliers, as well as its ability to invest resources to maintain its competitive position and expand its marketing and geographic reach.

These characteristics are important for a restaurant company because together they provide insight into the company's ability to drive revenue by increasing the average check amount, adding net new restaurants to its system, and attracting strong affiliates and franchisees. Greater scale, by extension, implies higher brand recognition and ability to expand – which is important given the competitive nature of the restaurant industry. Scale enables a company to gain better control of its purchasing and distribution channels, establish a more powerful market position, and leverage fixed costs. For smaller companies in regional markets, we consider absolute scale as well as regional scale.

How we assess it for the scorecard

We score this factor based on two quantitative subfactors, Revenue and Number of Systemwide Restaurants; and one qualitative subfactor: Revenue by Geographic Region.

Revenue

We use total reported revenue in billions of US dollars to measure scale, or to estimate it in the case of forward-looking expectations.

We include revenues generated from company-owned and operated restaurants in addition to royalty, franchise and initial fees from affiliates and franchisees, as well as rental income.

Number of Systemwide Restaurants

We use number of systemwide restaurants, including those owned and operated by the company as well as those franchised and affiliated.

Revenue by Geographic Region

We score this subfactor based on our qualitative assessment of the concentration of operating performance within a country, as well as the percentage derived from the various regions.

Factor: Business Profile (10% weight)

Why it matters

The business profile of a company is important because it greatly influences its ability to generate sustainable earnings and operating cash flow.

In the restaurant industry, companies that are characterized as having strong brand recognition often have a higher level of profitable sales by maintaining a steady improvement in traffic and allowing for sustainable pricing power. This tends to result in higher unit volumes and attract franchisees and affiliates that are more financially secure.

Maintaining or building a strong brand in the restaurant industry requires a company to provide a better guest experience or perceived value, which is defined in part by being convenient and also by having a varied menu that offers quality products at reasonable prices in an appealing environment. Overall, companies must create in customers the belief that they provide better value versus the competition.

Brand Diversity

Diversification through multiple brands focused on different concepts, each narrower in scope but allowing for a more in-depth product offering, help mitigate revenue volatility as consumer tastes change or evolve over time. A more diverse day part distribution not only increases total revenue generation but also helps operating margins by better leveraging a restaurant's fixed cost infrastructure.

Brand Strength

Brand strength is an important indicator of a restaurant's ability to attract and retain customers. Same-store sales and its various components provide indications of how consumers perceive a brand as well as the frequency and success of new products. They are a widely used indicator of a company's operating performance and its brand strength, although the components of same-store sales are often more telling than the aggregate. A company's ability to increase its average check amount while maintaining positive traffic is one indicator of a company's pricing power and brand strength. However, positive same-store sales can also mask an underlying deterioration of brand strength in situations where, for an extended period, the percentage increase in the average check amount exceeds the percentage decline in traffic. Although this relationship of positive average check and negative traffic can last for some time, a company's ability to raise prices is limited and can exacerbate traffic declines.

How we assess it for the scorecard

We score this factor based on two qualitative subfactors: Brand Diversity and Brand Strength.

Brand Diversity

We score this subfactor based on our qualitative assessment of the diversification of the revenue stream. We consider the number of brands or concepts, and the diversity of a company's product offering.

Brand Strength

We score this subfactor based on our qualitative assessment of same-store sales.

Factor: Profitability and Efficiency (10% weight)

Why it matters

A company's profitability matters because it supports the generation of sustainable cash flow and reflects the strength of the company's competitive position. Profitability also provides insights into a company's ability to manage key costs and pass them on to customers, which is an important indicator of the long-term viability of its business model. Strong profitability and operating efficiency can help a company to withstand economic and cyclical downturns that affect its competitive position and credit metrics.

How we assess it for the scorecard

We score this factor based on one quantitative subfactor: Return on Assets.[3]

Return on Assets

We use the ratio of net profit after tax before unusual items (NPATBUI) to average assets.

Factor: Leverage and Coverage (45% weight)

Why it matters

Leverage and debt service coverage provide insight into the level of risk a company is willing to undertake. The degree of leverage is an important indicator of a company's financial flexibility, its ability to invest in growth opportunities and its capacity to pay debt service.

How we assess it for the scorecard

We score this factor based on three quantitative subfactors: RCF/Debt, Debt/EBITDA and EBIT/Interest Expense.[4]

RCF / Debt

We use the ratio of retained cash flow (RCF) to total debt.

Debt / EBITDA

We use the ratio of total debt to earnings before interest, taxes, depreciation and amortization (EBITDA).

EBIT / Interest Expense

We use the ratio of earnings before interest and taxes (EBIT) to interest expense.

Factor: Financial Policy (15% weight)

Why it matters

Financial policy is an important rating determinant because it directly affects a company's debt levels, its strategic direction, the risk of adverse changes in financing and capital structure, and its overall credit quality. Financial policy encompasses the tolerance of management and the board for financial risk and their level of commitment to maintaining a strong credit profile.

Financial risk tolerance serves as a guidepost for investment and capital allocation. Liquidity management is an important aspect of financial policy and can provide insight into risk tolerance. Liquidity management is distinct from the level of liquidity, which we discuss in the "Other considerations" section.

How we assess it for the scorecard

We assess qualitatively a company's targeted credit profile or desired capital structure, and its ability to achieve its targets. We also consider a company's complexity and transparency, its track record of risk and liquidity management, and its public commitments and history of adhering to its commitments in these areas. We also consider management's operating performance and use of cash flow through different phases of economic and industry cycles, as well as how management responds to key events, such as changes in the credit markets and liquidity environment, legal actions, competitive challenges and regulatory pressures.

When considering event risks in the context of scoring financial policy, we assess the likelihood and potential effect of M&A or other types of balance-sheet-transforming events. We assess management's appetite for M&A activity, with a focus on the type of transactions (i.e., core competency or new business) and funding decisions. We also evaluate the frequency and materiality of acquisitions and previous financing choices. A history of debt-financed or credit-transforming acquisitions generally results in a lower score for this factor. We may also consider a company's willingness to sell itself and the negative repercussions that could result, such as through a leveraged buyout transaction.

We also consider a company's and its owners' past record of balancing shareholder returns and debtholders' interests. We are likely to take a negative view in scoring this factor when a company has a track record of favoring shareholder returns at the expense of debtholders.

Other considerations

Ratings may reflect other considerations that are not in the scorecard, usually because the credit importance of these considerations varies widely among companies in this sector or because they may be important only under certain circumstances or for a subset of companies. Such considerations can include financial controls and the quality of financial reporting; corporate legal structure; the quality and experience of management; assessments of corporate governance, as well as environmental and social considerations; exposure to uncertain licensing regimes; and possible government interference in some countries. Regulatory, litigation, liquidity,

technology and reputational risk, as well as changes to consumer and business spending patterns, competitor strategies and macroeconomic trends can also be considerations that affect ratings.

The following are examples of other considerations that may be reflected in our ratings and may cause ratings to be different from scorecard-indicated outcomes.

Environmental, social and governance considerations

Where environmental, social and governance (ESG) issues are meaningful for credit profiles, we incorporate them into our ratings analysis in a variety of ways in the application of our sector-specific methodologies. As one part of our overall credit analysis, we consider how ESG considerations could affect the qualitative and quantitative factors and subfactors in the scorecard.

Even where ESG considerations do not affect the measures in the scorecard, or where they cannot be quantified, we incorporate them into our overall analysis of credit drivers that are meaningful to the rating. As a result, we may incorporate these ESG risks qualitatively outside of the scorecard. As part of our ratings analysis, we may establish Issuer Profile Scores (IPSs), which indicate our opinion of the extent to which a given company is exposed to E, S and G risks (incorporating ESG-specific mitigants) or benefits from its exposure to E, S or G. The IPSs are inputs to credit ratings. For more information, see our methodology that describes our general principles for assessing ESG risks.[5]

Liquidity

Liquidity is an important rating consideration for all companies, and extremely weak liquidity can heavily affect ratings in many cases. Our liquidity assessment can be particularly important where there are significant short-term demands on a company's liquidity, for example, in highly seasonal operating environments where working capital needs must be considered. We form an opinion on likely short-term liquidity requirements from the perspective of both sources and uses of cash. For more information, see our liquidity cross-sector methodology.[6]

Regulatory considerations

Companies in this sector are subject to varying degrees of regulatory oversight. Effects of these regulations may entail limitations on operations, higher costs, and higher potential for technology disruptions and demand substitution. Regional differences in regulation, implementation or enforcement may advantage or disadvantage a particular company.

Our view of future regulatory requirements is important because it is the basis for our expectations of future financial metrics and of a company's ability to generate sufficient cash flow relative to its debt burden over the medium and longer term.

Management strategy

The quality of management is an important factor supporting a company's credit strength. Assessing the execution of business plans over time can be helpful in assessing management's business strategies, policies and philosophies, and in evaluating management performance relative to that of competitors and our projections. Management's track record of adhering to stated plans, commitments and guidelines provides insight into management's likely future performance, including in stressed situations.

Financial controls

We rely on the accuracy of audited financial statements to assign and monitor ratings in this sector. The quality of financial statements may be influenced by internal controls, including the proper tone at the top, centralized operations, and consistency in accounting policies and procedures. Auditors' reports on the effectiveness of internal controls, auditors' comments in financial reports and unusual restatements of financial statements or delays in regulatory filings may indicate weaknesses in internal controls.

Event risk

We also recognize the possibility that an unexpected event could cause a sudden and sharp decline in a company's fundamental creditworthiness, which may cause the assigned rating to be lower than the scorecard-indicated outcome. Event risks — which vary and can range from leveraged recapitalizations to sudden regulatory changes or liabilities from an accident — can overwhelm even a stable, well-capitalized company. Other types of event risks may include M&A, asset sales, spinoffs, shareholder distributions, litigation, natural disasters, pandemics, significant cybercrime events and geopolitical conflicts.

Excess cash balances

Some companies in this sector may maintain cash balances (meaning liquid short-term investments, as well as cash) that far exceed their operating needs. This excess cash can be an important credit consideration.

The amount of cash required to run a business can vary materially based on the regions of operation and the specific subsectors in which a company operates. As a result, we typically assess cash on the balance sheet qualitatively, taking into account a company's track record and its financial and liquidity policies. Cash holdings may be a material mitigant in our scenario analyses of potential acquisitions, share buybacks or special dividends. We assess possible uses of cash, especially for companies that may be acquisitive or under pressure to return cash to shareholders. By contrast, some companies maintain large cash holdings for long periods in excess of their operating and liquidity needs solely because of conservative financial policies, which provides a stronger indication of an enduring approach that will benefit creditors. Large cash holdings provide some buffers against credit deterioration, and we typically consider cash balances in our analysis of near-term liquidity sources and uses. See the "Liquidity" section for more information.

In accordance with our view of how a company will use its cash, we may use total debt or net debt in our forward-looking view of credit metrics for the Leverage and Coverage factor. We may also cite rating thresholds based on these metrics.

Additional metrics

The metrics included in the scorecard are those that are generally most important in assigning ratings to companies in this sector; however, we may use additional metrics to inform our analysis of specific companies. These additional metrics may be important in our forward-looking view of metrics in the scorecard or in other rating considerations.

Non-wholly owned subsidiaries

Some companies in this sector choose to dilute their equity stake in certain material subsidiaries, such as through an initial public offering or a sale to a private investor, which may, in some cases, negatively affect future financial flexibility. While such transactions may improve cash holdings on a one-off basis, selling minority interests in subsidiaries may have a negative effect on cash flow available to the parent company that consolidated financial statements may not fully reflect. For example, in the case of an equity stake reduction in a subsidiary down to 75%, in the parent's financial statements, the subsidiary's revenue and EBITDA would typically still be consolidated at the group level.

The presence of minority stakes can increase structural subordination and reduce the parent's share of dividend flows from a non-wholly owned subsidiary. While less frequent, the sale of a minority stake may have provisions that further limit the parent's financial flexibility, for instance restrictions on cash pooling with other members of the corporate family, limitations on dividends and distributions, or arms-length business requirements. Minority stakeholders may have seats on the board of the subsidiary.

In many cases, we consider the impact of non-wholly owned subsidiaries qualitatively. For example, we may find that an additional view of financial results, such as analyzing cash flow on a proportional consolidation basis, may be useful to inform our analysis. When equity dilution or structural subordination from non-wholly owned subsidiaries is material and negative, the credit impact is captured in our ratings.

For companies that hold material minority interest stakes (investor companies), consolidated funds from operations typically includes the dividends received from the equity-accounted entity, while none of its debt is consolidated. The investor company's cash flow may be subject to interruption if it is required for the equity-accounted entity's debt service, capital expenditures or other cash needs. The debt of the equity-accounted entity is structurally senior to the investor company's debt. When an equity-accounted entity has a material level of debt, we may estimate one or more of the investor company's financial metrics using proportional consolidation or another approach to inform our analysis of consolidated financials and capture the credit impact of these considerations in our ratings.

Parental support

Ownership can provide ratings lift for a particular company if it is owned by a highly rated owner or owners who view it to be of strategic importance. In our analysis of parental support, we consider whether the parent has the financial capacity and strategic incentives to provide support to the company in times of stress or financial need (e.g., a major capital investment or advantaged operating agreement), or if it has done so in the past. Conversely, if the parent puts a high dividend burden on the company, which in

turn reduces its flexibility, the ratings will reflect this risk. For more information, see our cross-sector methodology on assessing the impact of financial support from a parent, subsidiary or affiliate in the absence of a guarantee.[7]

Government-related issuers may receive ratings uplift from expected government support. However, for certain companies, government ownership can have a negative effect on the underlying Baseline Credit Assessment (BCA). For example, price controls, onerous taxation and high distributions can have a negative effect on a company's underlying credit profile. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[8]

Other institutional support

In some countries, large companies have received government or banking support in the event of financial difficulties because of their overall importance to the functioning of the economy. We consider whether group and banking relationships may provide support when systemically important companies encounter significant financial stress.

As a notable example, some of our corporate ratings in Japan incorporate the government or banking support we expect to be provided, based on historical experience, to large and systemically important companies. Over the years, this support has resulted in lower levels of default for Japanese companies than might otherwise have occurred.

Scorecard-indicated outcome

We arrive at a scorecard-indicated outcome by first mapping the outcome for each factor or subfactor to a numeric score, and then by producing an aggregate numeric score and the equivalent alphanumeric outcome.

Mapping scorecard factors to a numeric score

We map the outcome of each factor or subfactor in the scorecard to a broad Moody's rating category and to a corresponding numeric score. We score qualitative factors and subfactors based on the description by broad rating category in the scorecard. The following exhibit shows the numeric score that corresponds to each broad rating category.

Exhibit 3
Numeric scores for qualitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
1	3	6	9	12	15	18	20

Source: Moody's Ratings

We score quantitative factors and subfactors on a linear continuum. For each metric, the scorecard shows the range by broad rating category. We use the scale in the following exhibit and linear interpolation to convert the metric, based on its placement within the scorecard range, into a numeric score, which may be a fraction. As an illustrative example, if the Baa range for the ratio of revenue to interest is 50x to 100x, the numeric score for a company with a ratio of 99x – relatively strong within this range – would be closer to 7.5, while the numeric score for a company with a ratio of 51x – relatively weak within this range – would be closer to 10.5. In the scorecard footnotes, we define the endpoints of the line (i.e., the metric values that equate to the highest possible numeric score and the lowest possible numeric score).

Exhibit 4
Ranges of numeric scores for quantitative factors and subfactors

Aaa	Aa	A	Baa	Ba	B	Caa	Ca
0.5 - 1.5	1.5 - 4.5	4.5 - 7.5	7.5 - 10.5	10.5 - 13.5	13.5 - 16.5	16.5 - 19.5	19.5 - 20.5

Source: Moody's Ratings

Determining the overall scorecard-indicated outcome

We multiply the numeric score for each subfactor (or each factor, when the factor has no subfactors) by the weight of that subfactor (or factor) and sum the results to produce an aggregate numeric score. We map the aggregate numeric score back to a scorecard-indicated outcome using the ranges in the following exhibit.

Exhibit 5
Scorecard-indicated outcome

Aggregate numeric score	Scorecard-indicated outcome
x ≤ 1.5	Aaa
1.5 < x ≤ 2.5	Aa1
2.5 < x ≤ 3.5	Aa2
3.5 < x ≤ 4.5	Aa3
4.5 < x ≤ 5.5	A1
5.5 < x ≤ 6.5	A2
6.5 < x ≤ 7.5	A3
7.5 < x ≤ 8.5	Baa1
8.5 < x ≤ 9.5	Baa2
9.5 < x ≤ 10.5	Baa3
10.5 < x ≤ 11.5	Ba1
11.5 < x ≤ 12.5	Ba2
12.5 < x ≤ 13.5	Ba3
13.5 < x ≤ 14.5	B1
14.5 < x ≤ 15.5	B2
15.5 < x ≤ 16.5	B3
16.5 < x ≤ 17.5	Caa1
17.5 < x ≤ 18.5	Caa2
18.5 < x ≤ 19.5	Caa3
19.5 < x ≤ 20.5	Ca
x > 20.5	C

Source: Moody's Ratings

For example, a company with an aggregate numeric score of 11.7 would have a Ba2 scorecard-indicated outcome.

In general, the scorecard-indicated outcome is oriented to the senior unsecured rating for investment-grade companies and the corporate family rating (CFR) for speculative-grade companies. For companies that benefit from rating uplift from parental support, government ownership or other institutional support, we consider the underlying credit strength or BCA for comparison to the scorecard-indicated outcome. For more information on the BCA, see *Rating Symbols and Definitions* and our cross-sector methodology for government-related issuers.[9]

Assigning issuer-level and instrument-level ratings

After considering the scorecard-indicated outcome, other considerations and relevant cross-sector methodologies, we typically assign a senior unsecured rating for investment-grade companies and a CFR for speculative-grade companies. For government-related issuers, we assign a BCA.

We may notch individual debt instrument ratings upward or downward from the senior unsecured rating or the CFR to reflect our assessment of differences in expected loss related to an instrument's seniority level and collateral. For more information, see our cross-sector methodologies on loss given default for speculative-grade nonfinancial companies, notching corporate instrument ratings based on differences in security and priority of claim, and assigning short-term ratings.[10]

Key rating assumptions

For information about key rating assumptions that apply to methodologies generally, see *Rating Symbols and Definitions*.[11]

Limitations

In the preceding sections, we have discussed the scorecard factors and other considerations that may be important in assigning ratings. In this section, we discuss limitations that pertain to the scorecard and to the overall methodology.

Limitations of the scorecard

There are various reasons why scorecard-indicated outcomes may not map closely to assigned ratings.

For example, the scorecard may not fully capture credit loss and recovery considerations, which typically become more important as a company gets closer to default. The scorecard is also limited by its upper and lower bounds, causing scorecard-indicated outcomes to be less likely to align with ratings for companies at the upper and lower ends of the rating scale.

The weights of each scorecard factor and subfactor approximate their importance for rating decisions across the sector, but the importance of a particular factor may vary substantially based on an individual company's circumstances.

Factors outside the scorecard, including those discussed in the "Other considerations" section, may be important for ratings, and their relative importance may also vary from company to company. In addition, certain broad methodological considerations described in one or more cross-sector methodologies may be relevant to ratings in this sector.[12] These considerations include how sovereign credit quality affects nonsovereign issuers, the assessment of credit support from other entities, the relative ranking of different classes of debt and hybrid securities, and the assignment of short-term ratings.

We may use the scorecard over various historical or forward-looking periods. Furthermore, in our ratings we often incorporate directional views of risks and mitigants in a qualitative way.

General limitations of the methodology

This methodology does not include an exhaustive description of all the factors we may consider in assigning ratings in this sector. Companies in this sector may face new risks or new combinations of risks, and they may develop new strategies to mitigate risk. We seek to incorporate all material credit considerations in our ratings and to take the most forward-looking perspective that visibility into these risks and mitigants permits.

Ratings reflect our expectations for a company's future performance; however, as the forward horizon lengthens, uncertainty increases and the utility of precise estimates, as scorecard inputs or in other considerations, typically diminishes. Our forward-looking opinions are based on assumptions that may prove, in hindsight, to have been incorrect. Reasons for this could include unanticipated changes in any of the following: the macroeconomic environment, general financial market conditions, industry competition, disruptive technology, or regulatory and legal actions. In any case, predicting the future is subject to substantial uncertainty.

Moody's related publications

Credit ratings are primarily determined through the application of sector methodologies. Certain broad methodological considerations (described in one or more cross-sector methodologies) may also be relevant to the determination of credit ratings of issuers and instruments. A list of sector and cross-sector methodologies can be found here.

Data summarizing the historical robustness and predictive power of credit ratings can be found here.

For more information, see *Rating Symbols and Definitions*, which is available here.

Endnotes

1 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

2 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

3 For definitions of our most common metrics, see our cross-sector methodology that describes our financial statement adjustments for nonfinancial corporations. A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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5 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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7 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

8 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

9 Links to *Rating Symbols and Definitions* and to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

10 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

11 A link to *Rating Symbols and Definitions* can be found in the "Moody's related publications" section.

12 A link to a list of our sector and cross-sector methodologies can be found in the "Moody's related publications" section.

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